A logarithmic ceiling for sparse Euclidean grids in the cube

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Source problem

The source paper is Daniel J. Fresen, *Euclidean arrangements in Banach spaces*, arXiv:1309.5526. Problem 4 appears across pages 3–4 of the source PDF.

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The same dependence on $\|a\|_0$ can be achieved with the use of the Dvoretzky-Rogers factorization [4, 11, 39, 40, 43], but only on a subspace of proportional dimension. Note that in Corollary 3, logarithmic sparsity alone is not enough to guarantee bounded Euclidean distortion.

Problem 4 *Does there exist a universal constant $C > 0$ and a sequence $(\omega_n)_1^\infty$ with $\lim_{n\to\infty} \omega_n = \infty$ such that the following is true? For every $n \in \mathbb{N}$ and any real Banach*

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space $(X, \|\cdot\|)$ of dimension n, there is a basis $(e_i)_1^n$ for X such that for any $a \in \mathbb{R}^n$ with $\|a\|_0 \le \omega_n$,

$$\left(\sum_{i=1}^n a_i^2\right)^{1/2} \le \left\|\sum_{i=1}^n a_i e_i\right\| \le C\left(\sum_{i=1}^n a_i^2\right)^{1/2}$$

Can one take $\omega_n = c\log n$? Can one take $(e_i)_1^n$ to be orthonormal with respect to the John ellipsoid of B_X?

Problem 4 asks whether there are a universal constant $C > 0$ and a sequence $\omega_n \to \infty$ such that every real n-dimensional Banach space X admits a basis $(e_j)_{j=1}^n$ satisfying

$$\|a\|_2 \le \left\|\sum_{j=1}^n a_j e_j\right\|_X \le C\|a\|_2$$

whenever a is ω_n-sparse. It then asks whether one can take $\omega_n = c\log n$.

Proof Intuition

The cube gives an immediate ceiling on how large ω_n could ever be. Write a candidate basis of ℓ_∞^n as the columns of a matrix T. The upper sparse estimate says that every row of T, restricted to any k columns, has Euclidean norm at most C. Now fix any k columns and put random signs on them. For each row, Hoeffding's inequality says the signed average is smaller than 1 with probability at least $1 - 2\exp(-k/(2C^2))$. If k is much larger than $C^2 \log n$, a union bound over the n rows gives one sign choice whose image has ℓ_∞ norm less than 1, contradicting the required lower estimate.

Partial result. *Let $T : \mathbb{R}^n \to \ell_\infty^n$ be linear, let $C \geq 1$, and let $1 \leq k \leq n$. Suppose that for every k-sparse vector $a \in \mathbb{R}^n$,*

$$\|a\|_2 \leq \|Ta\|_\infty \leq C\|a\|_2.$$

Then

$$k \leq 2C^2 \log(2n).$$

Consequently, in Fresen's Problem 4 no affirmative answer can have $\omega_n/\log n \to \infty$; the logarithmic order asked about in the problem is the largest possible order, already because of the spaces ℓ_∞^n.

Proof. Let $r_i = (T_{ij})_{j=1}^n$ be the ith row of the matrix of T. The upper estimate implies that for every set $S \subset \{1, \ldots, n\}$ with $|S| \leq k$,

$$\|r_i|_S\|_2 = \sup_{\operatorname{supp} a \subseteq S,\ \|a\|_2 = 1} \left| \sum_{j \in S} T_{ij} a_j \right| \leq C$$

for every row i.

Fix one set S with $|S| = k$, and let $(\varepsilon_j)_{j \in S}$ be independent Rademacher signs. For row i set

$$Z_i = \frac{1}{\sqrt{k}} \sum_{j \in S} \varepsilon_j T_{ij}.$$

By Hoeffding's inequality, using $\sum_{j \in S} (T_{ij}/\sqrt{k})^2 \leq C^2/k$, we have

$$\mathbb{P}\{|Z_i| \geq 1\} \leq 2\exp\left(-\frac{k}{2C^2}\right).$$

Therefore

$$\mathbb{P}\left\{\max_{1 \leq i \leq n} |Z_i| \geq 1\right\} \leq 2n \exp\left(-\frac{k}{2C^2}\right).$$

If $k > 2C^2 \log(2n)$, the right-hand side is strictly less than 1. Hence there is a choice of signs such that all $|Z_i| < 1$.

For that choice, define $a_j = \varepsilon_j/\sqrt{k}$ for $j \in S$ and $a_j = 0$ otherwise. Then a is k-sparse and $\|a\|_2 = 1$, while

$$\|Ta\|_\infty = \max_i |Z_i| < 1.$$

This contradicts the assumed lower estimate. Thus $k \leq 2C^2 \log(2n)$.

If Problem 4 had a positive answer with some sequence $\omega_n/\log n \to \infty$, applying it to $X = \ell_\infty^n$ and to $k = \lfloor \omega_n \rfloor$ for large n would contradict the bound above. $\square$

Verification notes

The proof uses only Hoeffding's inequality and the literal two-sided sparse estimate from Problem 4. The basis assumption for ℓ_∞^n is encoded by writing the basis vectors as columns of T; invertibility is not used in the obstruction, so the argument applies a fortiori to bases.

A bounded novelty check was performed on 2026-06-14: the run indexes were searched for arXiv:1309.5526 and terms including "Euclidean grid", "ell_infty", "sparse", and "logarithmic"; external exact-phrase web queries for the source problem did not reveal a duplicate packet. Novelty confidence is low-to-moderate, because the result is close to standard entropy lower bounds for embedding Euclidean spaces into ℓ_∞^n.

Limitations

This is only a ceiling result. It does not construct bases with $k \simeq c \log n$, and it does not disprove the main existence question with some slower sequence $\omega_n \to \infty$. It shows that the second question in Problem 4, asking whether $\omega_n = c \log n$ is possible, is sharp in order if the first question has any positive solution.

Human-review recommendation

Check the constants in the Hoeffding step and decide whether to regard this as a new partial result or as a compact proof of standard folklore. In either case, it is useful duplicate memory for the main sparse-basis problem: future attempts should not aim for superlogarithmic sparsity.

References

[1] Daniel J. Fresen, *Euclidean arrangements in Banach spaces*, arXiv:1309.5526, 2013.